

# Dry Type Transformers

## General Information

Types QB, QMS, QL  
600 Volts and Below

## Section 10

### General Information

The complete family of transformers from GE provide quiet, reliable transformer operation.

All of the dry-type transformers through 1,000 kVA are UL listed under the requirements of Standard 5085 and 1561. In addition, each transformer meets the requirements of NEMA ST-20, 1992. Type IP, QB and QMS models are C-UL listed.

General-purpose transformers are rated 600 Volts and below for supplying appliance, lighting, and power loads from electrical distribution systems. Standard distribution voltages are 600, 480, and 240 Volts; standard load voltages are 480, 240, 208, and 120 Volts. The transformer is used to obtain the load voltage from the distribution voltage. Since no vaults are required for installation, these transformers can be located right at the load to provide the correct voltage for the application. This eliminates the need for long, costly, low-voltage feeders.

### Construction

#### Types QB and QMS

Core and coils are contained within a NEMA 3R nonventilated weatherproof enclosure. Type QB and QMS units feature encapsulated core and coils.

#### Type QL

Units are enclosed in a NEMA 2 drip-proof painted metal enclosure with natural draft ventilation. The core-and-coil assembly is mounted on rubber isolation pads to reduce noise. Weathershield kits are available for conversion to a NEMA 3R enclosure suitable for outdoor service. NEMA 2 and NEMA 3R stainless steel (Type 316) enclosures are available up to 150kVA. To specify a stainless steel enclosure for an aluminum-wound transformer, substitute the letter "S" in the fifth character of the GE product number. For example, 9T10A1004 changes to 9T10S1004. For copper-wound transformers, substitute the letter "Z" in the fifth character of the GE product number. For example, 9T10C1004 changes to 9T10Z1004. **All QL model product numbers begin with 9T7, 9T8, or 9T1.**

Transformer taps compensate for high or low line voltages. Most standard QL units rated 15kVA through 300kVA and with a primary voltage of 240V or higher have six available voltage taps – four 2.5% taps below the nominal tap and two 2.5% taps above the nominal tap. This arrangement provides a 15% range of tap voltage adjustment. Transformers rated 500kVA and higher have four available voltage taps – two 2.5% taps above the nominal tap and two 2.5% taps below the nominal tap.

### Temperature Class

Industry standards classify insulation systems in accordance with the rating system shown below.

| Insulation System Classification |                |            |               |
|----------------------------------|----------------|------------|---------------|
| Ambient                          | + Winding Rise | + Hot Spot | = Temp. Class |
| 40°C                             | 55°C           | 10°C       | 105°C         |
| 40°C                             | 80°C           | 30°C       | 150°C         |
| 40°C                             | 115°C          | 25°C       | 180°C         |
| 40°C                             | 150°C          | 30°C       | 220°C         |

All standard, general-purpose, GE transformers meet all applicable NEMA, ANSI, UL, and IEEE standards.

The design life of transformers having different insulation systems is the same, since the allowable temperature rise of an insulation material system is predicated on a specified life for all insulation. The lower temperature systems are designed for the same life as higher temperature systems.

### Sound Levels

All general-purpose transformers are as quiet, or quieter than required by NEMA ST-20. Average sound levels are warranted not to exceed the values listed for each load rating shown in the adjacent table. Sound characteristics vary between transformers of identical voltage and kVA rating. The range of variation may be 4 to 8 decibels.

These values apply only to specified test conditions because the characteristic of the installation can cause them to be higher under operating conditions. Where acoustical noise is deemed to be of unusual concern, proper steps should be taken during installation to minimize audible noise transmission.

#### Sound Levels (Decibels)<sup>1</sup> for 150°C Rise Models

| kVA       | Sound Levels |
|-----------|--------------|
| 0 - 9     | 40           |
| 10 - 50   | 45           |
| 51 - 150  | 50           |
| 151 - 300 | 55           |
| 301 - 500 | 60           |

<sup>1</sup>Measured per NEMA ST-20.

# Dry Type Transformers

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Types QB, QMS, QL  
600 Volts and Below

### Termination

Improved termination spacing and wiring compartment room gives greater flexibility in selecting various UL listed connectors for either copper or aluminum cable.

### Product Number Selection Instructions

1. Establish phase and frequency
2. Determine the primary voltage—the voltage presently available
3. Determine the secondary voltage—the voltage needed at the load
4. Determine the kVA load, allowing room for expansion
5. Using the facts determined in the four steps, locate the transformer model in the listings on the following pages.



Type QB, .050 kVA-3 kVA, Single-Phase



Type QMS, 5 kVA-25 kVA, Single-Phase



Type QL, 15 kVA-250 kVA, Single-Phase, DOE 2016 Efficiency,  
15 kVA-500 kVA, Three-Phase, DOE 2016 Efficiency

# Dry Type Transformers

## General Purpose

### Three-Phase

DOE 2016 Product Number Nomenclature

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|                  |  |                        |   |                          |     |              |     |                                          |  |
|------------------|--|------------------------|---|--------------------------|-----|--------------|-----|------------------------------------------|--|
| 9T               |  | 1                      | 0 | A                        | 100 | 1            | G31 |                                          |  |
| GE Standard      |  | 1 = QL DOE 2016 Design |   | Transformer Type         |     | kVA Rating   |     | Group Number                             |  |
| 0 = Standard, K1 |  | 1 = K13                |   | 2 = K20                  |     | 3 = 15kVA    |     | No Group Number Shown = 150°C            |  |
| 3 = K30          |  | 4 = K4                 |   | 7 = Guard II/Servicenter |     | 4 = 30kVA    |     | G02 = 150°C, -3dB                        |  |
| 8 = Spare Parts  |  | Coil Material          |   | A = Aluminum             |     | 5 = 45kVA    |     | G03 = 150°C, Electrostatic Shield†       |  |
| C = Copper       |  | Voltage Rating         |   | See Voltage Chart below  |     | 6 = 75kVA    |     | G04 = 150°C, Electrostatic Shield†, -3dB |  |
|                  |  |                        |   |                          |     | 7 = 112.5kVA |     | G05 = 150°C, -5dB                        |  |
|                  |  |                        |   |                          |     | 8 = 150kVA   |     | G06 = 150°C, Electrostatic Shield†, -5dB |  |
|                  |  |                        |   |                          |     | 9 = 225kVA   |     | G09 = 150°C, Non-STD kVA                 |  |
|                  |  |                        |   |                          |     |              |     | G10 = 150°C, Non-STD kVA                 |  |
|                  |  |                        |   |                          |     |              |     | G29 = 140°C, 40°C amb                    |  |
|                  |  |                        |   |                          |     |              |     | G30 = 140°C, 50°C amb                    |  |
|                  |  |                        |   |                          |     |              |     | G31 = 115°C                              |  |
|                  |  |                        |   |                          |     |              |     | G32 = 115°C, -3dB                        |  |
|                  |  |                        |   |                          |     |              |     | G33 = 115°C, Electrostatic Shield†       |  |
|                  |  |                        |   |                          |     |              |     | G34 = 115°C, Electrostatic Shield†, -3dB |  |
|                  |  |                        |   |                          |     |              |     | G35 = 115°C, -5dB                        |  |
|                  |  |                        |   |                          |     |              |     | G36 = 115°C, Electrostatic Shield†, -5dB |  |
|                  |  |                        |   |                          |     |              |     | G39 = 115°C, Non-STD kVA                 |  |
|                  |  |                        |   |                          |     |              |     | G40 = 115°C, Non-STD kVA                 |  |
|                  |  |                        |   |                          |     |              |     | G61 = 80°C                               |  |
|                  |  |                        |   |                          |     |              |     | G62 = 80°C, -3dB                         |  |
|                  |  |                        |   |                          |     |              |     | G63 = 80°C, Electrostatic Shield†        |  |
|                  |  |                        |   |                          |     |              |     | G64 = 80°C, Electrostatic Shield†, -3dB  |  |
|                  |  |                        |   |                          |     |              |     | G65 = 80°C, -5dB                         |  |
|                  |  |                        |   |                          |     |              |     | G66 = 80°C, Electrostatic Shield†, -5dB  |  |
|                  |  |                        |   |                          |     |              |     | G69 = 80°C, Non-STD kVA                  |  |
|                  |  |                        |   |                          |     |              |     | G70 = 80°C, Non-STD kVA                  |  |

†An Electrostatic Shield (also called a Guard I transformer) is a grounded, aluminum barrier between the primary and secondary winding to reduce electrical noise. All K-Factor transformers contain an Electrostatic Shield.

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### 3-Phase Common Voltages

| Series | Primary Voltage | Secondary Voltage | Series | Primary Voltage | Secondary Voltage | Series | Primary Voltage | Secondary Voltage |
|--------|-----------------|-------------------|--------|-----------------|-------------------|--------|-----------------|-------------------|
| 100    | 480             | 208Y/120          | 157    | 240             | 575               | 212    | 400             | 480Y/277          |
| 101    | 480             | 220               | 158    | 240             | 460Y/266          | 213    | 415             | 208Y/120          |
| 102    | 480             | 220Y/127          | 159    | 240             | 240Y/139          | 214    | 415             | 460               |
| 103    | 480             | 208               | 160    | 240             | 600               | 215    | 415             | 220Y/127          |
| 104    | 480             | 230Y/133          | 161    | 240             | 208Y/120          | 216    | 416             | 208Y/120          |
| 105    | 480             | 240Y/139          | 162    | 240             | 380               | 217    | 416             | 480Y/277          |
| 106    | 480             | 380               | 163    | 240             | 440               | 218    | 460             | 208Y/120          |
| 107    | 480             | 380Y/219          | 164    | 240             | 240/120           | 219    | 460             | 220               |
| 108    | 480             | 400Y/231          | 165    | 240             | 380Y/219          | 221    | 460             | 400Y/231          |
| 109    | 480             | 415Y/240          | 166    | 220             | 380Y/219          | 222    | 460             | 220Y/127          |
| 110    | 480             | 480               | 167    | 220             | 400Y/231          | 223    | 460             | 230               |
| 111    | 480             | 575               | 168    | 220             | 240Y/139          | 224    | 460             | 575Y/332          |
| 112    | 480             | 600               | 169    | 220             | 220               | 225    | 460             | 230Y/133          |
| 113    | 480             | 440Y/254          | 170    | 220             | 208Y/120          | 226    | 460             | 460Y/266          |
| 114    | 480             | 600Y/346          | 171    | 220             | 480Y/277          | 227    | 550             | 208Y/120          |
| 115    | 480             | 440               | 172    | 220             | 440Y/254          | 228    | 550             | 480Y/277          |
| 116    | 480             | 230/115           | 173    | 220             | 480/240           | 229    | 575             | 208Y/120          |
| 117    | 480             | 480Y/277          | 174    | 220             | 480               | 230    | 575             | 480Y/277          |
| 118    | 480             | 240/120           | 175    | 220             | 415Y/240          | 231    | 575             | 240Y/139          |
| 119    | 480             | 240               | 176    | 380             | 220Y/127          | 232    | 575             | 460               |
| 121    | 480             | 220/110           | 177    | 380             | 480               | 233    | 575             | 480               |
| 123    | 480             | 400               | 178    | 380             | 220               | 234    | 575             | 230Y/133          |
| 124    | 480             | 460               | 179    | 380             | 208Y/120          | 235    | 575             | 230               |
| 125    | 480             | 420               | 180    | 380             | 415Y/240          | 236    | 600             | 240/120           |
| 126    | 480             | 230               | 181    | 380             | 240/120           | 237    | 600             | 480               |
| 127    | 480             | 575Y/332          | 184    | 380             | 480Y/277          | 238    | 600             | 480Y/277          |
| 129    | 480             | 460Y/266          | 185    | 380             | 380Y/219          | 239    | 600             | 240               |
| 131    | 208             | 240               | 186    | 380             | 230Y/133          | 240    | 600             | 208Y/120          |
| 132    | 208             | 240/120           | 187    | 380             | 240               | 241    | 600             | 230Y/133          |
| 133    | 208             | 480               | 188    | 440             | 220Y/127          | 242    | 600             | 240Y/139          |
| 134    | 208             | 480Y/277          | 189    | 440             | 480               | 243    | 600             | 208               |
| 135    | 208             | 380Y/219          | 190    | 440             | 208Y/120          | 244    | 600             | 600Y/346          |
| 136    | 208             | 230               | 191    | 440             | 380               | 245    | 690             | 400Y/231          |
| 137    | 208             | 575               | 192    | 440             | 380Y/219          | 246    | 690             | 208Y/120          |
| 138    | 208             | 460               | 193    | 440             | 400Y/231          | 247    | 277             | 415Y/240          |
| 139    | 208             | 400Y/231          | 194    | 440             | 575Y/332          | 248    | 315             | 208Y/120          |
| 140    | 208             | 208               | 195    | 440             | 240/120           | 249    | 320             | 480Y/277          |
| 141    | 208             | 230Y/133          | 196    | 440             | 480Y/277          | 250    | 420             | 480Y/277          |
| 142    | 208             | 380               | 197    | 440             | 240               | 251    | 490             | 480Y/277          |
| 143    | 208             | 220/110           | 198    | 230             | 460               | 252    | 500             | 480Y/277          |
| 144    | 208             | 220Y/127          | 199    | 230             | 400Y/231          |        |                 |                   |
| 145    | 208             | 208Y/120          | 200    | 230             | 400               |        |                 |                   |
| 146    | 208             | 400               | 201    | 230             | 480Y/277          |        |                 |                   |
| 147    | 208             | 315               | 202    | 230             | 208Y/120          |        |                 |                   |
| 148    | 208             | 600               | 203    | 230             | 480               |        |                 |                   |
| 149    | 208             | 460Y/266          | 204    | 400             | 230Y/133          |        |                 |                   |
| 150    | 208             | 220               | 205    | 400             | 380Y/219          |        |                 |                   |
| 151    | 208             | 230/115           | 206    | 400             | 480               |        |                 |                   |
| 152    | 208             | 415Y/240          | 207    | 400             | 220Y/127          |        |                 |                   |
| 153    | 240             | 480Y/277          | 209    | 400             | 400Y/231          |        |                 |                   |
| 154    | 240             | 480               | 210    | 400             | 208Y/120          |        |                 |                   |
| 155    | 240             | 400Y/231          | 211    | 400             | 208Y/120          |        |                 |                   |

# Dry Type Transformers

## General Purpose Encapsulated

### Single-Phase NEMA 3R

## Section 10

.050 - 3 kVA Indoor/Outdoor Type QB UL Listed C-UL Listed

| Input Voltage              | Output Voltage | kVA   | Frequency (Hz) | Taps         | Wiring Diagram No. <sup>1</sup> | Weight (Lbs) | Frame Size | Product Number |
|----------------------------|----------------|-------|----------------|--------------|---------------------------------|--------------|------------|----------------|
| 240 x 480 Volts, 480 Volts | 120/240 Volts  | 0.05  | 60 Hz          | No Taps      | 1                               | 6            | 6100       | 9T51B0002      |
| 240 x 480 Volts            | 120/240 Volts  | 0.05  | 50 Hz          | No Taps      | 1                               | 6            | 6100       | 9T51B0502      |
| 240 x 480 Volts, 480 Volts | 120/240 Volts  | 0.075 | 60 Hz          | No Taps      | 1                               | 6            | 6200       | 9T51B0003      |
| 240 x 480 Volts            | 120/240 Volts  | 0.075 | 50 Hz          | No Taps      | 1                               | 6            | 6200       | 9T51B0503      |
| 240 x 480 Volts, 480 Volts | 120/240 Volts  | 0.1   | 60 Hz          | No Taps      | 1                               | 6            | 6200       | 9T51B0004      |
| 240 x 480 Volts            | 120/240 Volts  | 0.1   | 50 Hz          | No Taps      | 1                               | 6            | 6200       | 9T51B0504      |
| 240 x 480 Volts, 480 Volts | 120/240 Volts  | 0.15  | 60 Hz          | No Taps      | 1                               | 10           | 8175       | 9T51B0005      |
| 240 x 480 Volts            | 120/240 Volts  | 0.15  | 50 Hz          | No Taps      | 1                               | 10           | 8175       | 9T51B0505      |
| 240 x 480 Volts, 480 Volts | 120/240 Volts  | 0.25  | 60 Hz          | No Taps      | 1                               | 10           | 8175       | 9T51B0007      |
| 240 x 480 Volts            | 120/240 Volts  | 0.25  | 50 Hz          | No Taps      | 1                               | 10           | 8200       | 9T51B0507      |
| 240 x 480 Volts, 480 Volts | 120/240 Volts  | 0.5   | 60 Hz          | No Taps      | 1                               | 16           | 10200      | 9T51B0008      |
| 240 x 480 Volts            | 120/240 Volts  | 0.5   | 50 Hz          | No Taps      | 1                               | 20           | 10225      | 9T51B0508      |
| 240 x 480 Volts, 480 Volts | 120/240 Volts  | 0.75  | 60 Hz          | No Taps      | 1                               | 25           | 12200      | 9T51B0009      |
| 240 x 480 Volts            | 120/240 Volts  | 0.75  | 50 Hz          | No Taps      | 1                               | 25           | 12225      | 9T51B0509      |
| 240 x 480 Volts, 480 Volts | 120/240 Volts  | 1     | 60 Hz          | No Taps      | 1                               | 25           | 12225      | 9T51B0010      |
| 240 x 480 Volts            | 120/240 Volts  | 1     | 50 Hz          | No Taps      | 1                               | 30           | 12275      | 9T51B0510      |
| 240 x 480 Volts, 480 Volts | 120/240 Volts  | 1.5   | 60 Hz          | No Taps      | 1                               | 40           | 14200      | 9T51B0011      |
| 240 x 480 Volts            | 120/240 Volts  | 1.5   | 50 Hz          | No Taps      | 1                               | 40           | 14225      | 9T51B0511      |
| 240 x 480 Volts, 480 Volts | 120/240 Volts  | 2     | 60 Hz          | No Taps      | 1                               | 45           | 14250      | 9T51B0012      |
| 240 x 480 Volts            | 120/240 Volts  | 2     | 50 Hz          | No Taps      | 1                               | 50           | 14300      | 9T51B0512      |
| 240 x 480 Volts, 480 Volts | 120/240 Volts  | 3     | 60 Hz          | No Taps      | 1                               | 55           | 14350      | 9T51B0013      |
| 240 x 480 Volts            | 120/240 Volts  | 3     | 50 Hz          | No Taps      | 1                               | 60           | 14400      | 9T51B0513      |
| 480 Volts                  | 120/240 Volts  | 0.5   | 50 Hz          | (-2 5.0%)    | 2                               | 20           | 10225      | 9T51B0548      |
| 480 Volts                  | 120/240 Volts  | 0.75  | 50 Hz          | (-2 5.0%)    | 2                               | 25           | 12200      | 9T51B0549      |
| 480 Volts                  | 120/240 Volts  | 1     | 60 Hz          | (-2 5.0%)    | 2                               | 25           | 12225      | 9T51B0050      |
| 480 Volts                  | 120/240 Volts  | 1     | 50 Hz          | (-2 5.0%)    | 2                               | 30           | 12275      | 9T51B0550      |
| 480 Volts                  | 120/240 Volts  | 1.5   | 60 Hz          | (-2 5.0%)    | 2                               | 40           | 14200      | 9T51B0051      |
| 480 Volts                  | 120/240 Volts  | 1.5   | 50 Hz          | (-2 5.0%)    | 2                               | 40           | 14225      | 9T51B0551      |
| 480 Volts                  | 120/240 Volts  | 2     | 60 Hz          | (-2 5.0%)    | 2                               | 45           | 14250      | 9T51B0052      |
| 480 Volts                  | 120/240 Volts  | 2     | 50 Hz          | (-2 5.0%)    | 2                               | 50           | 14300      | 9T51B0552      |
| 480 Volts                  | 120/240 Volts  | 3     | 60 Hz          | (-2 5.0%)    | 2                               | 55           | 14350      | 9T51B0053      |
| 480 Volts                  | 120/240 Volts  | 3     | 60 Hz          | (+2,-2 2.5%) | 3                               | 55           | 14350      | 9T51B0135      |
| 480 Volts                  | 120/240 Volts  | 3     | 50 Hz          | (-2 5.0%)    | 2                               | 60           | 14400      | 9T51B0553      |
| 600 Volts                  | 120/240 Volts  | 0.05  | 60 Hz          | No Taps      | 5                               | 6            | 6100       | 9T51B0082      |
| 600 Volts                  | 120/240 Volts  | 0.075 | 60 Hz          | No Taps      | 5                               | 6            | 6200       | 9T51B0083      |
| 600 Volts                  | 120/240 Volts  | 0.1   | 60 Hz          | No Taps      | 5                               | 6            | 6200       | 9T51B0084      |
| 600 Volts                  | 120/240 Volts  | 0.1   | 50 Hz          | No Taps      | 5                               | 6            | 6200       | 9T51B0584      |
| 600 Volts                  | 120/240 Volts  | 0.15  | 60 Hz          | No Taps      | 5                               | 10           | 8175       | 9T51B0085      |
| 600 Volts                  | 120/240 Volts  | 0.25  | 60 Hz          | No Taps      | 5                               | 10           | 8175       | 9T51B0087      |
| 600 Volts                  | 120/240 Volts  | 0.25  | 50 Hz          | No Taps      | 5                               | 10           | 8200       | 9T51B0587      |
| 600 Volts                  | 120/240 Volts  | 0.5   | 60 Hz          | No Taps      | 5                               | 16           | 10200      | 9T51B0088      |
| 600 Volts                  | 120/240 Volts  | 0.5   | 50 Hz          | (-2 5.0%)    | 2                               | 20           | 10225      | 9T51B0568      |
| 600 Volts                  | 120/240 Volts  | 0.75  | 60 Hz          | No Taps      | 5                               | 25           | 12200      | 9T51B0089      |
| 600 Volts                  | 120/240 Volts  | 0.75  | 50 Hz          | (-2 5.0%)    | 2                               | 25           | 12200      | 9T51B0569      |
| 600 Volts                  | 120/240 Volts  | 1     | 60 Hz          | (-2 5.0%)    | 2                               | 25           | 12225      | 9T51B0070      |
| 600 Volts                  | 120/240 Volts  | 1     | 60 Hz          | No Taps      | 5                               | 25           | 12225      | 9T51B0090      |
| 600 Volts                  | 120/240 Volts  | 1     | 50 Hz          | (-2 5.0%)    | 2                               | 30           | 12275      | 9T51B0570      |
| 600 Volts                  | 120/240 Volts  | 1.5   | 60 Hz          | (-2 5.0%)    | 2                               | 40           | 14200      | 9T51B0071      |
| 600 Volts                  | 120/240 Volts  | 1.5   | 60 Hz          | No Taps      | 5                               | 40           | 14200      | 9T51B0091      |
| 600 Volts                  | 120/240 Volts  | 1.5   | 50 Hz          | (-2 5.0%)    | 2                               | 40           | 14225      | 9T51B0571      |
| 600 Volts                  | 120/240 Volts  | 2     | 60 Hz          | (-2 5.0%)    | 2                               | 45           | 14250      | 9T51B0072      |
| 600 Volts                  | 120/240 Volts  | 2     | 60 Hz          | No Taps      | 5                               | 45           | 14250      | 9T51B0092      |
| 600 Volts                  | 120/240 Volts  | 2     | 50 Hz          | (-2 5.0%)    | 2                               | 50           | 14300      | 9T51B0572      |
| 600 Volts                  | 120/240 Volts  | 2     | 50 Hz          | No Taps      |                                 | 50           | 14300      | 9T51B0592      |
| 600 Volts                  | 120/240 Volts  | 3     | 60 Hz          | (-2 5.0%)    | 2                               | 55           | 14350      | 9T51B0073      |
| 600 Volts                  | 120/240 Volts  | 3     | 60 Hz          | No Taps      | 5                               | 55           | 14350      | 9T51B0093      |
| 600 Volts                  | 120/240 Volts  | 3     | 50 Hz          | (-2 5.0%)    | 2                               | 60           | 14400      | 9T51B0573      |
| 600 Volts                  | 120/240 Volts  | 3     | 50 Hz          | No Taps      |                                 | 60           | 14400      | 9T51B0593      |

<sup>1</sup>See page 10-45 for wiring diagrams.

# Dry Type Transformers

## Buck-Boost

### Encapsulated

For Bucking and Boosting Voltage

## Section 10

### Product Description

Buck boost transformers are small, single-phase, dry type distribution transformers designed and shipped as insulating/isolating transformers. They have a dual voltage primary and a dual voltage secondary. These transformers can be connected for a wide range of voltage combinations. The most common use is to buck (lower) or boost (raise) the supply voltage a small amount, usually 5 to 27%. Buck boost transformers comply with NEC Article 210-9, Exception 1, when field connected as an autotransformer.

GE bucking and boosting transformers provide an economical and convenient means for bucking or boosting voltage, usually no more than  $\pm 20\%$  on single- and three-phase circuits. They are compact, relatively light in weight, and can be easily installed for indoor or outdoor service.

Buck-boost transformers are employed primarily for boosting single- and three-phase circuits by connecting them as autotransformers. When connected as an autotransformer, only the low-voltage, high-current capacity secondary windings are required to carry the load. Because this load is only transformed over a small change in voltage, the buck-boost transformer can handle loads many times its nameplate kVA rating.

The transformers with series-multiple 12/24, 24/48, or 16/32 Volt secondary windings are suitable for a wide variety of applications. Two or more units can be used in various combinations to obtain many other special voltages. (For fluctuating voltage conditions, refer to Power Conditioning Equipment Products section starting on page 10-36).

### Advantages

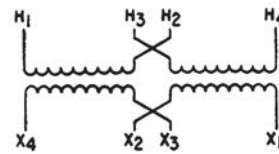
- Efficient insulating materials permit compact size and light weight
- Dual voltage primary and dual voltage secondary for maximum versatility
- Large, front-accessible wiring compartment permits fast, easy wiring
- Convenient conduit knockouts located on side, bottom and back of wiring compartment
- GE Buck-Boost Transformer Selector makes selection fast and easy
- Many GE buck-boost transformers fit competitor mounting footprints
- Indoor or outdoor service

### Key Features

- Convenient and least expensive method of matching line voltage with equipment voltage
- More efficient than equivalent isolation transformers
- Ability to handle loads up to 20 times nameplate rating when connected as an autotransformer
- Ideal for changing line voltages by small amounts
- Primary voltages include 120V, 240V and 480V
- Secondary voltages include 12V, 16V, 24V, 32V, 48V
- UL and cUL Listed



Indoor/Outdoor Type QB Transformer; Single-Phase



Wiring Diagram for Low Voltage Loads

- Qualified to the seismic requirements of IEEE-693-1997 and IBC-2003
- ABS (American Bureau of Shipping) Type Approved

### Applications

- International voltage adaptation
- Commercial and industrial air conditioning
- Heating systems
- Induction motors
- Voltage line drop correction
- Landscape lighting
- Low-voltage lighting
- Marine and Offshore - ABS Classed Vessels

Efficient operation of electrical equipment requires that line voltage be at or near the nameplate rating of the equipment. In order to match available line voltage (whether it be too high or low) with equipment voltage, buck-boost transformers provide the most convenient and least expensive method.

Do not use buck-boost transformers to solve a fluctuating voltage problem. They should be used to compensate for high- or low-voltage conditions only when the available line voltage is reasonably constant.